

BitCoin's roller coaster: Systemic risk and Market Sentiment

Panagiota Makrichoriti*, Georgios Moratis †

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Abstract

This study investigates the determinants of Bitcoin exchange rate over the period Feb. 2013 – Apr. 2016 within a VAR framework. We are mainly driven by the narrow knowledge of this high volatile cryptocurrency's behavior. We ask the question how and to what extent the overall credit and liquidity conditions affect Bitcoin. A novel feature of this paper is its important implications for the investors in order to understand Bitcoin's determinants. Also, the empirical investigation is conducted by means of VAR model that permits the analysis of multi-directional relationships in a dynamic context. Our findings indicate that investor sentiment is the only important Bitcoin determinant among other factors which capture the overall economic situation, while spillover effects measured by the spillover matrix introduced by Diebold and Yilmaz (2009) indicate that Bitcoin appears to be a “lonely boat” that is mostly affected by “idiosyncratic” factors. Hence, it is very important for investors to gain knowledge on Bitcoin's special characteristics, while paying attention to the investors sentiment.

Keywords: *Cryptocurrency, Bitcoin, Systemic risk, Vector Autoregressive models*

JEL classification: *C35, E44, F31, G15*

*Athens University of Economics and Business, 76, Patission Str., GR10434 Athens-Greece. Email: pegy_makrichoriti@hotmail.gr

†Corresponding author: Athens University of Economics and Business, 76, Patission Str., GR10434 Athens-Greece. Email: g.moratis@yahoo.com (corresponding author)

1 Introduction

As Arthur C. Clarke¹ said *“two possibilities exist: either we are alone in the Universe or we are not. Both are equally terrifying.”* To some extent, this quote has implications for international finance, as spillover effects and interdependence among markets transformed the financial landscape. Bitcoin exchange rates exhibit somewhat complicated dynamics and its exchange rate is not well aligned with those of other currencies. At a first glance, Bitcoin variation seems to be released by financial markets’ “leash”, as its fluctuations were going through the roof. However, the apparent opportunities that Bitcoin offers may include risks, whose nature and importance are not well understood, and affect everyone who uses Bitcoin as a currency or trades it as an asset. For instance, Graham et al. (2005) find that most executives feel they make an appropriate choice, when sacrificing economic value for smooth earnings or for hitting a target. So, how could they ever choose such a volatile vehicle, when at the same time they do not have broad knowledge of its determinants?

The biggest concern on Bitcoin is its high volatility and its sensitive price. A recent illustrative example of intense currency fluctuation is the case of euro. More precisely, until May 2008, Euro gained almost 100% against the US dollar, while during the global financial crisis it lost more than 30% of its value. However, the fluctuation on Bitcoin’s price during the period November 2013 to January 2015 is unprecedented. Bitcoin’s appreciation was almost zero in 2009; it rose to 1,163 (to USD) in November 2013 and then declined to 167 (to USD). In this setting therefore an interesting topic for research is whether Bitcoin’s intense behavior may be attributed to the general market conditions or to “idiosyncratic” factors.

The present study uses monthly data covering the period from February 2013 to April 2016. We ask the following three questions. How have the global and Euro-zone systemic risk factors affected Bitcoin? Does Bitcoin affect the general credit and liquidity conditions? Finally, does the market sentiment have an effect on Bitcoin? In this setting, the questions asked are answered within a VAR framework, on the basis of the results obtained from Forecast Error Variance Decomposition (FEVD) analyses, Impulse Response Functions (IRFs) and spillover matrix analysis (Diebold and Yilmaz, 2009), giving the opportunity to investigate not only the market anticipation of Bitcoin but also to investigate whether Bitcoin has an effect on the overall economic condition in global markets.

The results suggest that Bitcoin is not related to the overall economic situation and it appears to be like a “lonely boat” that is mostly affected by investors sentiment and “idiosyncratic” factors. The main contribution of the essay is its important implications for the investors in order to understand the driving forces of Bitcoin. The second contribution stems from the modeling

¹Sir Arthur Charles Clarke (16 December 1917 - 19 March 2008) was a British science fiction writer

approach that has been chosen and from the fact that this is the first study that attempts to investigate the effect of market conditions on Bitcoin’s behavior. This methodology does not consider the changes on Bitcoin as an exogenous event, but it treats it as a variable that is being determined endogenously within a dynamic process. Another important contribution is that this is the first study that attempts to investigate the effect of market sentiment on a currency. The relevant literature is narrow and limited to studies that are related to sovereign risk. For instance, Georgoutsos and Miyakis (2012) put an emphasis on the importance of uncertainty and investment confidence conditions in the pricing of sovereign debt.

The rest of the paper is structured as follows. In section two, the testing methodology is presented along with the definition of the data used in the analysis. In section three, the results are presented as well as the evidence from a series of robustness tests. Finally, section four, concludes with a summary of the main contributions of the paper.

2 Methodology & Data

The general starting point for the empirical analysis is the reduced-form presentations of a linear VAR:

$$y_t = C + A_1 y_{t-1} + \dots + A_p y_{t-p} + \varepsilon_t$$

with $t = 1, \dots, T$ and T being the sample size, y is an $n \times 1$ vector of endogenous variables and C is an $n \times 1$ vector of regression constants and ε is an $n \times 1$ vector of reduced-form shocks with assumed distribution $\varepsilon_t \sim i.i.d.N(0, \Omega)$. The coefficients C and A are estimated by running ordinary least squares regressions equation by equation, and the variance-covariance matrix is estimated from the sample residuals as $\Omega = (1/T) \sum_{t=1}^T \varepsilon_t \varepsilon_t'$. The data have a monthly frequency, spanning the period Feb 2013 – April 2016. We use Bitcoin to US dollar exchange rate (XBT/USD) and it is important to mention that we receive qualitatively similar results when replacing US dollar with euro (XBT/EUR). The global default risk conditions are represented by the CDX index, the world capital markets’ “fear” condition by the VIX-CBOE volatility index and the world liquidity conditions by the TED variable which is defined as the difference between interest rates on interbank loans and short-term U.S. government debt. The corresponding variables for Euro-zone are the iTraxx for Europe, the Vstox and the 3 month Euribor-Eonia spread². The variables we choose to express global financial risk conditions have been widely used in related studies as control variables (see, among others, De Santis 2014, Aizenman et al. 2013, Ang and

² Bitcoin’s exchange rate, CDX, iTraxx, Vstox and VIX are transformed into arithmetic returns, while TED spread and Euribor Eonia spread are used in first differences and the market sentiment is transformed into log differences (the importance of the results remains the same when MCSI is employed in levels).

Longstaff 2013). Finally, the MCSI is a market sentiment indicator published by the University of Michigan and is typically employed in empirical studies to measure expectations and consumer optimism and pessimism, or as a predictor of asset returns (see, among others, Barsky and Sims, 2012; Lemmon and Portniaguina, 2006).

In the last part of our analysis we have applied the methodology of Diebold and Yilmaz (2009, 2012) in order to measure the intensity of the spillover effects from shocks to each one of the variables. This methodology relies on Generalized Forecast Error Variance Decomposition (GFEVD), obtained as shown in Pesaran and Shin (1998), which permits the calculation of a spillover matrix. The rows of this matrix show the individual impact, over a number of periods, of a shock to one variable (impulse variable) on each one of the other variables (response variables) in the system as well as the total sum of the impacts on all the variables (sum out). In a similar way the columns of the matrix show the impact received of an individual variable from shocks on each one of the other variables as well as the total received impact (total in). This matrix facilitates the identification of the variables that are responsible for the instability caused in the system, offering supplementary evidence on the main drivers of Bitcoin. <Insert Table 1 and Table 2>

3 Empirical Results

We start with the presentation of unit-root tests for all the variables in our VAR model. We have applied Phillips-Perron test (1998) for unit-root, which is an Augmented Dickey-Fuller (ADF) type test on time-series data and Table 3 shows that not in a single case we were not able to reject the null hypothesis for the existence of a unit root <Insert Table 3>. We have conducted test on the stability properties of the estimated VAR model (Lutkepohl, 2005) and we use the LM test for autocorrelation (results are available upon request).

In Table 4 we present the coefficients from the VAR model for the chosen period. In this model, the systemic risk factor, the market sentiment and the Bitcoin’s exchange rate, are treated as endogenous. The results suggest that Bitcoin is not affected by any of the variables included in the systemic risk factor, but it is affected strongly and negatively by the market sentiment. This result is illustrative, showing that Bitcoin’s performance is “immune” to the general market conditions but it is affected by the market expectations <Insert Table 4>. In table 5, we present the evidence from variance decomposition analysis. The empirical evidence move to the same direction, showing clearly that only global market expectations play a role in the explanation of Bitcoin’s volatility. More precisely, a shock to the US investors sentiment explains a 26.87 percent of Bitcoin’s variation, while a shock in global liquidity is able to explain a 13.81 percent of Bitcoin’s variation. It seems that Bitcoin’s behavior is mostly determined by “idiosyncratic”

factors and market expectations, as they explain together a 62.05 percent of Bitcoin’s variation <Insert Table 5>. In table 6, we present results from the spillover matrix which tracks the variables that cause the instability in the system. Once more, it is clear that results still hold, in the sense that Bitcoin can not be characterized as a shock receiver, neither as a shock sender as the relevant percentage is close to zero (0.22). This percentage is by far the smallest in the system. It is illustrative that while VIX (23.68), TED spread (2.54), Euribor Eonia spread (7.87) and iTraxx (1.23) are net senders of shocks, none of these shocks seem to be delivered to Bitcoin <Insert Table 6>. Finally, figure 1 presents the results from the impulse response functions, which confirm the results found in previous analysis, as they show that the response of Bitcoin is never significantly different than zero during the first two steps, but it is significantly different than zero in the cases of the response to market sentiment and to its own shock. Also, the IRF reveals a negative relationship between US investor sentiment and the Bitcoin, suggesting that if investors are confident that the country’s economy will be strong, they will be more likely to buy the country’s assets, and probably not the Bitcoin. If investors are not confident that a country’s economy will be strong, they will be less likely to buy the country’s assets, pushing down the value of that country’s currency, and more likely to buy the Bitcoin.<Insert Figure 1>

4 Conclusions

Bitcoin exchange rate volatility affects everybody who uses Bitcoin as a currency or trades it as an asset. Since Bitcoin’s fluctuations were going through the roof and knowledge about its determinants is narrow, the whole topic deserves attention. Is high volatility helpful for Bitcoin or not? The answer to this question could not be black or white, as it clearly depends on the perspective from which you look at it. Volatility means risk, and this potentially means bigger returns. Nevertheless, whether you are an investor, a “miner”, a merchant or a an average user, it is very important to understand the determinants of Bitcoin.

To the best of our knowledge, this is the first study that empirically investigates the determinants of Bitcoin within a VAR setting. We use a systemic risk factor and a market expectations index to decode Bitcoin’s behavior. All in all, we could say that Bitcoin’s behavior cannot be explained by variables that capture the overall economic situation and it appears to be like a “lonely boat” which is affected only by its own characteristics and investors sentiment. The general picture has important implications for Bitcoin users in order to understand Bitcoin’s “nature”. Hence, it is very important for investor to gain knowledge on Bitcoin’s special characteristics, while taking under consideration the effect of market sentiment.

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Table 1: Variables Description

<i>Variable</i>	<i>Description</i>
<i>CDX</i>	The family of tradable CDS indices covering North America and emerging markets
<i>VIX</i>	The volatility index of S&P 500
<i>TED</i>	The difference between the interest rates on interbank loans and on short-term US government debt
<i>MCSI</i>	Market confidence indicator published by the University of Michigan
<i>EurEon</i>	The difference between the “Euro Interbank Offered rate” and the “Euro Overnight Index Average”
<i>iTraxx</i>	iTraxx Europe main index (125 investment grade companies, all sectors)
<i>Vstoxx</i>	The volatility index of EURO STOXX 50
<i>Bitcoin</i>	Bitcoin exchange rate (XBT/USD)

Source: Thomson - Reuters Datastream, Bloomberg

Table 2: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
<i>CDX</i>	39	.002	.102	-.268	.164
<i>VIX</i>	39	.038	.304	-.384	1.345
<i>TED</i>	39	.004	.028	-.07	.07
<i>MCSI</i>	39	.005	.037	-.059	.113
<i>EurEon</i>	39	-.001	.108	-.402	.314
<i>iTraxx</i>	39	-.002	.130	-.271	.269
<i>Vstoxx</i>	39	.031	.214	-.402	.610
<i>Bitcoin</i>	39	.193	.792	-.343	4.510

Notes: The Table presents descriptive statistics for the following variables: The family of tradable CDS indices covering North America and emerging markets (denoted as CDX), the volatility index of S&P500 (denoted as VIX), the difference between the interest rates on interbank loans and on short-term U.S. government debt (denoted as TED), the difference between the “Euro Interbank Offered rate” and the “Euro Overnight Index Average”(denoted as EurEon), the iTraxx Europe main index (denoted as iTraxx), the volatility index of EURO STOXX 50 (denoted as VSTOXX), the Market Confidence Market Indicator published by the University of Michigan (denoted as MCSI) and the the Bitcoin exchange rate in USD (denoted as Bitcoin). All of the time series are transformed to ensure stationarity; TED spread and Euribor-Eonia spread are used in first differences, MCSI in log differences, while CDX, VIX, iTraxx, VSTOXX and Bitcoin are used in growth rate. All data are monthly and obtained from DataStream International and Bloomberg. The sample covers the period between February 2013 and April 2016.

Table 3: Phillips-Perron test for Unit-root test (*3 lags*)

Variables	Test Statistic Z(t)	Interpollated Dickey-Fuller			MacKinnon p-value
		1% Crit. Val.	5% Crit. Val.	10% Crit. Val.	
<i>CDX</i>	-8.427	-3.662	-2.964	-2.614	0.000
<i>VIX</i>	-9.011	-3.662	-2.964	-2.614	0.000
<i>TED</i>	-8.736	-3.662	-2.964	-2.614	0.000
<i>MCSI</i>	-9.351	-3.662	-2.964	-2.614	0.000
<i>EurEon</i>	-11.405	-3.662	-2.964	-2.614	0.000
<i>iTraxx</i>	-8.025	-3.662	-2.964	-2.614	0.000
<i>Vstox</i>	-11.464	-3.662	-2.964	-2.614	0.000
<i>Bitcoin</i>	-5.542	-3.662	-2.964	-2.614	0.000

Table 4: Coefficients of VAR system

Variables	<i>CDX</i>	<i>VIX</i>	<i>TED</i>	<i>MCSI</i>	<i>EurEon</i>	<i>iTraxx</i>	<i>Vstox</i>	<i>Bit</i>
<i>L.CDX</i>	0.073 (0.383)	2.127* (1.266)	0.194** (0.084)	-0.312** (0.158)	0.362 (0.354)	0.186 (0.433)	0.648 (0.806)	-3.031 (2.987)
<i>L.VIX</i>	0.075 (0.098)	-0.393 (0.326)	-0.017 (0.021)	0.062 (0.040)	-0.123 (0.091)	0.150 (0.112)	0.071 (0.208)	0.711 (0.771)
<i>L.TED</i>	0.591 (0.503)	2.021 (1.859)	-0.236* (0.125)	0.158 (0.232)	0.726 (0.520)	1.687*** (0.636)	1.258 (1.183)	-3.332 (4.387)
<i>L.MCSI</i>	0.306 (0.420)	0.839 (1.385)	0.137 (0.092)	-0.525*** (0.173)	-0.300 (0.387)	-0.159 (0.474)	1.007 (0.881)	-9.625*** (3.268)
<i>L.EurEon</i>	0.338** (0.159)	0.704 (0.526)	0.083** (0.035)	-0.014 (0.065)	-0.596*** (0.147)	0.570*** (0.180)	0.613* (0.355)	-1.777 (1.241)
<i>L.iTraxx</i>	0.041 (0.305)	-1.226 (1.008)	-0.049 (0.067)	0.159 (0.126)	-0.207 (0.282)	0.256 (0.345)	0.479 (0.042)	-0.531 (2.379)
<i>L.Vstox</i>	-0.397** (0.173)	-0.614 (0.572)	-0.039 (0.038)	-0.075 (0.071)	0.187 (0.160)	-0.714*** (0.196)	-1.241*** (0.364)	-0.047 (1.351)
<i>L.Bit</i>	-0.043 (0.018)	-0.059 (0.061)	0.001 (0.004)	0.011 (0.007)	-0.021 (0.017)	-0.051** (0.021)	-0.037 (0.039)	0.118 (0.145)

Notes: The Table reports coefficients of regressing the dependent variables on lags of the independent variables. Sample period February 2013 – April 2016.

* denotes significance at 10% level

** denotes significance at 5% level

*** denotes significance at 1% level

Table 5: Forecast Error Variance Decomposition

Response	Impulse Variables							
	<i>CDX</i>	<i>VIX</i>	<i>TED</i>	<i>MCSI</i>	<i>EurEon</i>	<i>iTraxx</i>	<i>Vstoxx</i>	<i>Bit</i>
<i>CDX</i>	55.84	5.73	3.48	15.44	2.6	5.04	5.63	6.2
<i>VIX</i>	46.82	26.78	1.4	5.92	1.83	5.63	5.37	6.2
<i>TED</i>	1.44	24.83	52.25	4.73	6.38	4.12	4.18	2.04
<i>MCSI</i>	5.07	4.93	2.96	75.96	1.12	1.84	3.86	4.23
<i>EurEon</i>	10.98	23.43	9.1	9.5	37.94	1.66	3.67	3.68
<i>iTraxx</i>	39	7.56	7.28	12.82	4.19	11.72	11.26	6.13
<i>Vstoxx</i>	36.76	18.92	1.67	4.38	3.47	8.19	21.39	5.15
<i>Bit</i>	4.3	7.2	13.81	26.87	6.62	1.95	4.02	35.18

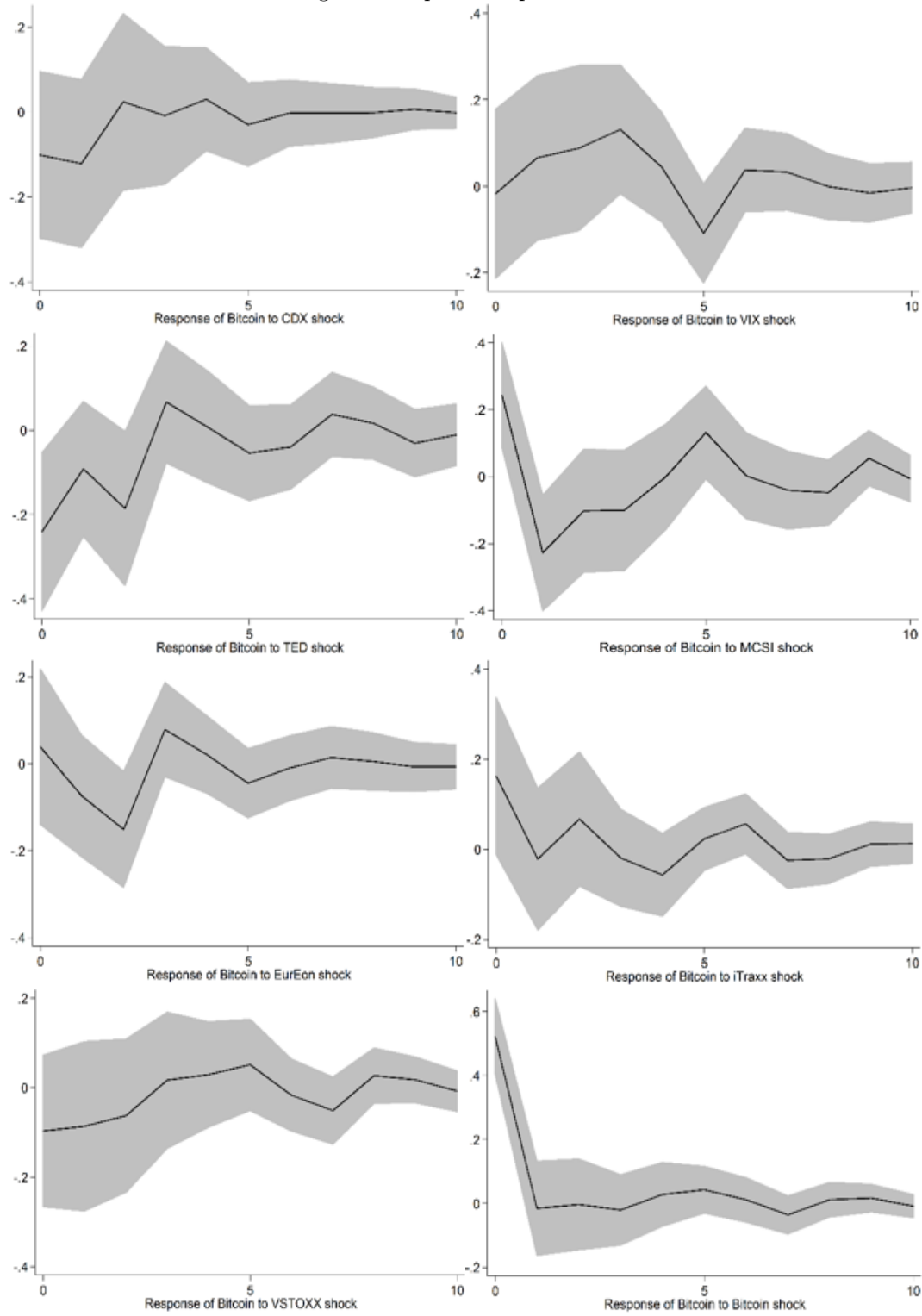
Notes: The tables report the fraction (in percentage points) of the 10-months ahead forecast error variance of each variable. The Cholesky ordering used is $CDX \rightarrow VIX \rightarrow TED \rightarrow MCSI \rightarrow EurEon \rightarrow iTraxx \rightarrow VSTOXX \rightarrow Bitcoin$.

Table 6: Spillover Matrix

Response/ Impulse	<i>CDX</i>	<i>VIX</i>	<i>TED</i>	<i>MCSI</i>	<i>EurEon</i>	<i>iTraxx</i>	<i>Vstoxx</i>	<i>Bitcoin</i>	<i>Sum Out</i>
<i>CDX</i>	55.84	38.36	4.49	16.88	9.36	46.42	35.56	9.32	130.39
<i>VIX</i>	46.82	71.62	6.42	5.94	3.96	40.62	57.75	9.03	170.54
<i>TED</i>	1.44	8.53	66.72	4.79	13.16	4.65	7.16	12.86	52.59
<i>MCSI</i>	5.07	3.05	2.38	79.96	4.87	4.76	5.39	12.21	37.73
<i>EurEon</i>	10.98	10.14	6.51	12.57	61.57	6.34	7.11	9.56	63.51
<i>iTraxx</i>	39	28.39	9.72	12.74	10.86	47.17	37.54	11.7	149.95
<i>Vstoxx</i>	36.76	52.91	6.57	4.4	5.03	41.57	71.15	5.4	152.64
<i>Bitcoin</i>	4.3	5.48	13.96	28.98	8.4	4.36	4.82	54.89	70.3
<i>Sum In</i>	144.37	146.86	50.05	86.3	55.64	148.72	155.33	70.08	857
<i>Net</i>	-13.98	23.68	2.54	-48.57	7.87	1.23	-2.69	0.22	-3

Notes: Variables in the first column are the impulse origin. Variables on the top row are the respondents to the shock. Values in the matrix represent the average cumulated spillover effect. The cumulative impact is bound between 0 and 1. A value of 0.5 means that the response variable will be impacted in the same direction with an intensity of 50% the initial unexpected shock in the impulse variable. In the last column we have the aggregated impact sent (Sum OUT) by each row variable and on the bottom row the aggregated spillover received (Sum IN) by each column variable. The bottom-right cell (in bold) shows total spillover in the system (by dividing this value to the total number of non-diagonal cells i.e. 7x6 we obtain the contagion index. The “Net” row represents the net spillover of each variable (Net Spillover=Sum OUT-Sum IN).

Figure 1: Impulse Response Functions



Note: Errors are 5% on each side generated by Monte-Carlo with 200 reps